THE HNS ORIGIN TRILOGY

Brain Architecture as the Foundation of Trustworthy AI

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A Note on Method and Evidence

This book documents the foundational programme of Human Natural Structure (HNS), situating it within the bodies of theory it draws upon. Each claim is connected to its intellectual ancestry and to the proponents who originated the underlying ideas: a dedicated chapter (IV, Intellectual Lineage) consolidates this lineage, worked examples (Appendix A) make the failure taxonomy concrete, a chapter on empirical status (VI) distinguishes carefully between what HNS demonstrably provides and what remains a research hypothesis, and a glossary (Appendix B) fixes the terminology.

Three registers are kept distinct throughout: *(i) established results* of the cited literatures; *(ii) the HNS design programme*, i.e., engineering proposals that follow from them; and *(iii) open conjectures* that motivate the programme but are not yet demonstrated. The programme's strongest claims — the geometric multiplication of error reduction, the brain–AI isomorphism, the zero-incidence figure — belong to (ii) or (iii) and are flagged as such. A framework candid about its evidentiary boundaries is harder to dismiss than one that is not.

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Preface — The Reliability Crisis in Artificial Intelligence

Artificial intelligence stands at a turning point. Large language models generate text indistinguishable from human writing, assist in research, advise on medical and legal questions, and participate in decisions affecting millions. Yet they fail in ways human experts do not: they assert falsehoods with confidence, lose track of the question, draw causal conclusions from mere correlations, let a metaphor displace a fact, and drift without signalling from the topic at hand (Ji et al., 2023; Ouyang et al., 2022).

These failures are not incidental bugs but structural properties of the architecture. A language model samples, at each step, from a high-dimensional distribution over the vocabulary (Vaswani et al., 2017). Output is fluent because high-probability sequences are coherent — but fluency and validity are independent. A sentence can be statistically natural and causally groundless. Probability does not confer structure.

The dominant response has been to scale (Ouyang et al., 2022; Bai et al., 2022). Scaling produces capability but not structural reliability: a larger model makes fewer surface errors yet remains open to the same deep failure classes — Layer Jump, Category Ambiguity, Scope Drift, Metaphor Contamination, Unsupported Causality — because these arise from a property scaling does not change: the model runs on a single computational axis with no independent mechanism to verify whether its outputs are structurally valid, causally grounded, or contextually coherent.

The question is where reliable cognition comes from — not in AI but in the brain. That intelligence is the cooperation of many semi-independent processes, monitored and integrated, rather than one engine, is old: it animates Minsky's Society of Mind (Minsky, 1986), Global Workspace Theory (Baars, 1988; Dehaene & Naccache, 2001), and the dual-process account of reasoning (Stanovich & West, 2000; Kahneman, 2011). If we can understand how the brain achieves reliability on a stochastic substrate, we may engineer that mechanism. This is the programme of Human Natural Structure (HNS).

About This Book

This volume collects the three foundational papers as a unified trilogy, proceeding from the neuroscientific theory (Chapter I), through the engineering architecture (Chapter II), to the structural correspondence (Chapter III). It continues with a chapter on intellectual lineage (IV), the EVA specification (V), a candid account of empirical status (VI), and the governance roadmap (VII), with worked examples and a glossary as appendices.

Structure of the Trilogy

No.	Title	Domain	Core contribution
I	The Multi-Axis Verification Process of the Human Brain	Computational neuroscience	Formalises MAV as the brain's mechanism for reliable cognition from stochastic parts
II	Human Natural Structure	AI safety & alignment	Engineers MAV into LLM pipelines via HNS-36/144/864 and SMS-6
III	Brain and AI with HNS: Correspondence	Cognitive systems	Maps biological cognition to HNS-aligned AI, level by level

Chapter I — The Multi-Axis Verification Process of the Human Brain

A brain-inspired architecture for geometric error-correction in stochastic cognitive systems.

1.1 The Reliability Paradox

A prefrontal pyramidal neuron integrates thousands of stochastic synaptic inputs; vesicle-release probability per spike falls between 0.1 and 0.9, and firing cannot be made deterministic by averaging (Faisal, Selen & Wolpert, 2008; Destexhe, Rudolph & Paré, 2003). Yet the surgeon's scalpel lands within a fraction of a millimetre, the mathematician holds a proof across twenty minutes, the listener resolves an ambiguous sentence in under 200 ms.

This is the reliability paradox: how does a system of noisy components produce outputs of such fidelity? Population coding (Pouget, Dayan & Zemel, 2000) reduces variance by 1/N but cannot explain why the brain's systematic errors — confabulation, categorical confusion, delusional causality — are patterned and situation-specific. A complementary mechanism must be at work. That reliable cognition is an assembly of cooperating specialists, monitored and integrated, is the thesis of the Society of Mind (Minsky, 1986) and of Global Workspace Theory (Baars, 1988; Dehaene & Changeux, 2011). MAV adds a specific geometric account of why such an assembly is reliable.

1.2 The MAV Hypothesis

MAV proposes that the brain's primary error-suppression mechanism is not within-axis averaging but the geometric intersection of three mutually (approximately) orthogonal verification processes: a Stochastic/Generative Axis, a Structural/Causal Axis, and a Grounded/Executive Axis. Their intersection identifies the output satisfying all three constraint classes at once (Miller & Cohen, 2001; Friston, 2010).

This output is X*invariant*, stable in that a perturbation along one axis does not displace it, because the other two re-anchor it. The intuition that independent constraints multiply reliability has a long pedigree: the Condorcet Jury Theorem (Condorcet, 1785) shows independent better-than-chance judges aggregate toward certainty, and ensemble learning and error-correcting output codes formalise the same in machine learning (Hansen & Salamon, 1990; Dietterich & Bakiri, 1995; Dietterich, 2000). The decisive condition is always independence: correlated constraints do not multiply.

In information-geometric form, when three constraint surfaces are mutually orthogonal (the Fisher information matrix inducing zero correlation among their defining parameters), the intersection volume approximately factorises:

$$Vol(A_1 \cap A_2 \cap A_3) \approx Vol(A_1) \cdot Vol(A_2) \cdot Vol(A_3) \,/\, Vol(\mathcal{M})^2$$

The framing rests on information geometry (Amari & Nagaoka, 2000), built on the Fisher–Rao metric (Rao, 1945) and Čencov's uniqueness theorem (Čencov, 1972). *The factorisation holds exactly only under idealised orthogonality, which real systems do not satisfy; it is a motivating model, not an established theorem about brains or LLMs, and its formal demonstration on neural representations is open (Chapter VI).*

1.3 The Three Axes

Axis 1 — Stochastic / Generative

Implemented by cortical association areas (DMN, lateral temporal and parietal cortex), Axis 1 generates high-prior candidates within tens of milliseconds: $A_1 = \{x \mid P(x \mid \text{context}, \theta) \geq \varepsilon_1\}$. Its volume is large; many valid and invalid outputs share high prior probability. It supplies generativity, not validity — the analogue of dual-process "System 1" (Stanovich & West, 2000; Evans & Stanovich, 2013; Kahneman, 2011).

Axis 2 — Structural / Causal

Implemented by the prefrontal–hippocampal network, Axis 2 defines permissible structure independent of frequency: dorsolateral PFC maintains hierarchical causal dependencies (Miller & Cohen, 2001); the hippocampus binds episodic sequences and enforces category boundaries (Eichenbaum, 2004). It enforces category-boundary, causal-ordering, and hierarchical-scope conditions: $A_2 = \{x \mid \forall i: c_i(x)=\text{TRUE}\}$. The standard for causal claims is that of causal-inference theory (Pearl, 2009); binding features without illusory conjunction is the binding problem (Treisman & Gelade, 1980; von der Malsburg, 1981).

Axis 3 — Grounded / Executive

Implemented by the sensorimotor system, ACC, and the predictive-coding hierarchy, Axis 3 is the "reality-friction interface." The ACC monitors prediction–outcome conflict (Botvinick et al., 2001): $A_3 = \{x \mid G(x)=1\}$, G a real-time grounding function. Its backbone is predictive coding (Rao & Ballard, 1999), the Bayesian brain (Knill & Pouget, 2004; Clark, 2013), and the free-energy principle (Friston, 2010); its engineering ancestry is cybernetics (Wiener, 1948), the TOTE unit (Miller, Galanter & Pribram, 1960), and perceptual control theory (Powers, 1973).

1.4 The Orthogonality Conditions

MAV's power depends on the axes being approximately orthogonal:

- **Axis 1 ⊥ Axis 2:** statistical frequency is independent of causal validity; θ is logically independent of the causal set C.
- **Axis 2 ⊥ Axis 3:** structural constraints live over category/causal space, grounding over the current environmental state.
- **Axis 1 ⊥ Axis 3:** priors are over the training distribution, which can diverge from the present context.

Perfect orthogonality is idealised; residual correlation defines the *irreducible error floor* — errors persisting with all axes active, plausibly the universal cognitive biases (Tversky & Kahneman, 1974). Measuring this residual via Fisher-information metrics is an open problem, not a settled result.

1.5 Failure Modes and Clinical Correlates

MAV predicts that selective axis disruption yields distinct failure modes — testable against neuropsychiatric phenomenology, and mirrored in the failures of single-axis LLMs. Worked examples of each (in the AI case) appear in Appendix A.

Failure type	Axis	Definition	HNS suppression	Clinical correlate
Layer Jump	2	Traversal of meaning layers without bridging	HNS-36 + SMS-6	Formal thought disorder
Category Ambiguity	2	Cognitive/emotional/intentional categories conflated	HNS-144 + SMS-5	Semantic dementia;

				psychosis
Scope Drift	3	Departure from intended scope without signalling	SMS-2 + SMS-3	Frontal-lobe confabulation
Metaphor Contamination	3	Figurative expression displaces coordinates	SMS-1 + SMS-6	Concretisation in schizotypy
Unsupported Causality	2+3	Causal claim without mechanistic grounding	HNS-864 + SMS-4	Delusional reasoning

1.6 Implications for Artificial Intelligence

Any system on a single computational axis is structurally limited in the reliability it can reach, however tightly that axis is optimised: the error volume of a single-axis strategy has a lower bound independent of its tightness. This is the hypothesis — and it is a hypothesis — underlying the engineering of Chapter II (Vaswani et al., 2017; Ji et al., 2023).

Chapter II — Human Natural Structure

Implementing the brain's Multi-Axis Verification process for advanced AI alignment.

2.1 The Structural Inadequacy of Current Alignment

RLHF, prompt guardrails, and RAG are single-axis interventions: they modify or supplement the generative axis without an independent, orthogonal constraint structure (Ouyang et al., 2022; Bai et al., 2022; Lewis et al., 2020). RLHF encodes alignment in the same substrate that generates errors; guardrails catch surface patterns, not structural incoherence; RAG supplies facts but can misapply them causally. What is missing is the engineering of Axes 2 and 3 as independent constraint planes intersecting the output in real time.

Architecturally, HNS is neuro-symbolic: a statistical generator coupled to a symbolic verification layer (d'Avila Garcez & Lamb, 2023; Marcus, 2020; Kautz, 2022). Its distinctive commitment is that the symbolic layer is *external and auditable*, not learned into the weights.

2.2 The HNS Architecture

HNS is a structural operating system instantiating the MAV constraint planes within the inference pipeline of a deployed LLM, modifying neither weights nor architecture (Hara, 2026a). It has two independent components — the Cellular Matrix (Axis 2: HNS-36/144/864) and the Social Meta Structure Specification (Axis 3: SMS-6) — integrated by the hns36 runtime, a pre-decode interception layer computing $A_1 \cap A_2 \cap A_3$ at every step.

2.3 Axis 2 — The HNS Cellular Matrix

HNS-36: Base Coordinate System

A 6×6 grid: six rows are natural causal layers from physical substrate to macro-social systems; six columns are abstract categories (entity, process, relation, state, transformation, meta-level). The ambition that a small category set tiles representable structure has roots in Aristotle's Categories and Kant's categories of understanding (Kant, 1781), and modern descendants in upper ontologies — DOLCE (Gangemi et al., 2002), BFO (Smith, 2015), SUMO (Niles & Pease, 2001) — and frame semantics (Fillmore, 1976). HNS-36 screens for Layer Jump: crossing non-adjacent cells without a transitional proposition is flagged ($c_i(x)$=FALSE) and attenuated.

HNS-144: Logical Relation Extension

Each cell expands along four logical-relation dimensions — subjective/objective, static/dynamic, local/global, necessary/contingent — yielding 144 cells, targeting Category Ambiguity by enforcing concept-boundary conditions.

HNS-864: Precision Causal Audit

Each 144-cell subdivides by six verification invariants, yielding 864 cells at the atomic-proposition level. A causal token passes only if its chain is traceable to a valid cell-transition path — the admissibility standard of causal inference (Pearl, 2009), with the communicative dimension drawing on Habermas (1984).

2.4 Axis 3 — The HNS-SMS-6 Protocol

Six-layer dynamic grounding: G(x)=1 only if compatible with all six layers; G(x)=0 triggers correction. The six layers extend grounding from body to civilisation — a scaling of embodied cognition (Barsalou, 2008) and the symbol-grounding requirement (Harnad, 1990).

Layer	Domain	Grounding function	Failure mode	Brain analogue
SMS-1	Somatic	Physical/ergonomic coherence with environment	Metaphor Contam.	Sensorimotor feedback
SMS-2	Protocol	Communicative alignment; intelligibility	Scope Drift	ACC conflict monitor
SMS-3	Ecosystem	Organisational & collaborative scope	Scope Drift	Parietal context
SMS-4	Economic	Resource/information equivalence; value	Unsupported Caus.	Orbitofrontal value
SMS-5	Governance	Conformance with external norms	Category Ambig.	PFC rule enforcement
SMS-6	Universal	Global optimality; no systemic violation	Layer Jump	Prefrontal coherence

2.5 The hns36 Runtime Engine

The runtime sits between hidden-state computation and decoding. Candidates failing the 36/144/864 filters are attenuated; candidates with $G(x)=0$ under SMS-6 are blocked; only mass in $A_1 \cap A_2 \cap A_3$ survives. Each check is logged with its coordinate or SMS layer. Functionally it is a runtime monitor in the sense of runtime verification (Leucker & Schallhart, 2009): a synchronous observer checking an executing system against a formal specification. Pseudocode appears in Chapter V.

2.6 Empirical Validation (Preliminary)

The runtime was integrated with a 70B open-source base LLM (no fine-tuning), compared against baseline and guardrail configurations over 50-turn sequences (150 turns total), annotated by two reviewers.

Failure type	Baseline	Guardrail	HNS
Layer Jump	18.2%	12.4%	0.0%
Category Ambiguity	14.7%	11.8%	0.0%
Scope Drift	21.5%	15.3%	0.0%
Metaphor Contamination	9.3%	8.6%	0.0%
Unsupported Causality	16.8%	13.1%	0.0%
Overall (any type)	43.1%	28.9%	0.0%

These are the reported PoC figures. *They must be read with the caveats of Chapter VI:* the failure types are framework-internal, the demonstration is single-model and not independently graded, and a zero-incidence result on framework-internal categories is therefore weaker than it appears. The property it most securely establishes is auditability of corrections, not a model-independent elimination of error.

2.7 Auditability

Every correction is attributable to a specific cell or SMS layer, logged at the time of correction. Because correction is mediated by explicit coordinates rather than opaque probability redistribution, the audit trail is a structural consequence of the architecture. The precise, defensible claim: HNS produces an externally checkable record of how an output was constrained. Whether that record is also complete and decision-relevant for a given governance purpose is an empirical question for the auditors who consume it (Chapters V, VI).

2.8 Beyond the Scaling Paradigm

Scaling is an Axis-1 strategy: it tightens a single hypersurface, which has an error-volume lower bound independent of its tightness. HNS pursues reliability through the intersection of lightweight Axes 2 and 3. If the orthogonality argument holds, the gains differ qualitatively from scaling's, at a fraction of the cost — and establishing that 'if' is the work of the programme, not a settled conclusion.

Chapter III — Brain and AI with HNS: A Correspondence

3.1 The Structural Correspondence

It is tempting to describe the mapping as a structural *isomorphism* — a structure-preserving bijection. That word must be qualified: what is demonstrated is a *structured analogy* — a level-by-level correspondence of functional roles — not a proven bijection. The proper framework is Marr's three levels (Marr, 1982): systems may correspond at the computational and algorithmic levels while differing at the implementational level. HNS is claimed to correspond to the brain at the computational/algorithmic levels; no claim is made about neural implementation.

Axis	Human brain	AI with HNS	Primary role	Suppresses
1	Cortical association (DMN, temporal, parietal)	LLM core (next-token)	Statistical generation	—
2	Prefrontal–hippocampal network	HNS-36/144/864	Structural & causal verification	Layer Jump, Category Ambiguity, Unsup. Causality
3	Sensorimotor system – ACC	HNS-SMS-6	Grounding & contextual verification	Scope Drift, Metaphor Contam.

3.2 Component-Level Mapping

Human brain	AI with HNS
Default Mode Network (DMN)	LLM generative core
Prefrontal–hippocampal network	HNS-36 / 144 / 864
Sensorimotor system – ACC	HNS-SMS-6

Each correspondence is a claim of functional role, defended at Marr's computational level (Friston, 2010; Destexhe et al., 2003).

3.3 Implications for AI Governance

The opacity of neural systems is the central regulatory challenge: alignment encodes corrections across billions of parameters with no legible account. HNS grounds verification in explicit coordinates corresponding, by the structured analogy, to identifiable functional roles — addressing requirements across frameworks (ISO/IEC

42001:2023; NIST, 2023; European Commission, 2024), developed in Chapter VII.

Chapter IV — Intellectual Lineage and Related Work

This chapter situates HNS within the theories it draws upon, names their proponents, and states precisely where HNS extends, recombines, or merely re-describes existing work. Novelty is clearer, not weaker, when lineage is explicit.

4.1 Architecture of mind: cooperating processes and a global workspace

Minsky's Society of Mind (1986) argued that intelligence emerges from many simple, specialised agents with no central controller. Global Workspace Theory made the integrating mechanism precise: specialised processors compete for access to a limited-capacity 'workspace' whose contents are broadcast back to all processors, producing the coherence and reportability we call consciousness (Baars, 1988); Dehaene and Naccache (2001) and Dehaene and Changeux (2011) gave this a neuronal formulation in terms of long-range cortical 'ignition.' MAV can be read as a GWT-style architecture in which the integrating workspace is the intersection of three constraint axes. HNS adopts the architectural intuition and contributes a geometric account of why intersection improves reliability — a question GWT itself leaves open.

4.2 Two systems of reasoning

Dual-process theory distinguishes a fast, automatic, associative system (Type 1) from a slow, effortful, rule-governed one (Type 2). Wason and Evans introduced the distinction empirically; Stanovich and West (2000) formalised individual differences; Evans and Stanovich (2013) defended the modern two-process account; Kahneman (2011) synthesised it for a general audience, and Tversky and Kahneman (1974) catalogued the biases that arise when Type 1 goes unchecked. In MAV terms, Axis 1 is Type 1 and Axes 2–3 supply the Type-2 check; the universal biases are the residue surviving imperfect orthogonality. HNS is, in one sentence, an attempt to give a Type-1-only language model an explicit, externalised Type-2 layer.

4.3 Metacognition and monitoring

Metacognition is cognition about cognition — the monitoring and control of one's own processing. Flavell (1979) named it; Nelson and Narens (1990) gave the canonical monitoring–control framework (object-level vs. meta-level, with monitoring flowing up and control flowing down); Fleming and Lau (2014) formalised the measurement of metacognitive accuracy. HNS's audit log is an externalised metacognitive trace: the monitoring decisions that in humans remain largely implicit are, in HNS, made explicit and machine-readable. This is the precise sense in which HNS 'knows what it corrected.'

4.4 Prediction, grounding, and control

Axis 3 rests on three convergent literatures. Predictive coding (Rao & Ballard, 1999) models perception as hierarchical prediction-error minimisation, each level predicting the level below and passing up only the residual error; the Bayesian brain (Knill & Pouget, 2004; Clark, 2013) casts this as approximate Bayesian inference; the free-energy principle (Friston, 2010) unifies perception and action as the minimisation of variational free energy. The symbol-grounding problem (Harnad, 1990) names the requirement that symbols connect to the world rather than circularly to other symbols, and embodied cognition (Barsalou, 2008) locates meaning in sensorimotor simulation. The engineering ancestry is cybernetic: Wiener's feedback control (1948), Ashby's law of requisite variety (1956), the TOTE unit (Miller, Galanter & Pribram, 1960), and perceptual control theory (Powers, 1973), in which behaviour is the control of perception against a reference. SMS-6's contribution is to extend grounding outward from the body to organisational, economic, governance, and civilisational scales.

4.5 Geometry of reliability

Information geometry treats families of probability distributions as differentiable manifolds with the Fisher information as a Riemannian metric (Amari & Nagaoka, 2000), a structure grounded in the Fisher–Rao metric (Rao, 1945) and shown to be the unique invariant metric by Čencov (1972). The complementary principle — that aggregating independent estimators multiplies reliability — is Condorcet's (1785) and reappears in ensemble learning (Hansen & Salamon, 1990; Dietterich, 2000) and error-correcting output codes (Dietterich & Bakiri, 1995). HNS's specific move is to apply this not to many homogeneous voters but to three heterogeneous, orthogonal constraint classes, and to read their intersection geometrically.

4.6 Categories, causation, and binding

The 6×6 category scheme descends philosophically from Aristotle's Categories and Kant's table of the categories of understanding (Kant, 1781), and computationally from upper ontologies — DOLCE's distinctions among endurants, perdurants, qualities, and abstracts (Gangemi et al., 2002), BFO's continuant/occurrent split (Smith, 2015), and SUMO (Niles & Pease, 2001) — and from frame semantics (Fillmore, 1976). The standard for admissible causal claims is Pearl's (2009): a causal assertion must be licensed by an admissible path in a causal model, distinguishing causation from the do-operator's intervention from mere association. The requirement that features bind into coherent objects without illusory conjunction is the binding problem (Treisman & Gelade, 1980; von der Malsburg, 1981).

4.7 Symbolic verification of statistical systems

HNS belongs to neuro-symbolic AI — the integration of neural learning with symbolic reasoning — surveyed by d'Avila Garcez and Lamb (2023), argued for by Marcus (2020), and taxonomised by Kautz (2022). Its verification layer is, in formal-methods terms, a runtime monitor: runtime verification (Leucker & Schallhart, 2009) studies exactly such synchronous observers that check an executing system against a temporal-logic specification and react to violations. Its failure taxonomy sits alongside other taxonomies of generation error (Maynez et al., 2020; Ji et al., 2023; Huang et al., 2023). HNS's claim to novelty rests not on any single element but on combining them into an external, auditable, brain-motivated verification layer for deployed LLMs.

4.8 Lineage at a glance

HNS element	Drawn from	Proponent(s)
Multi-axis cooperation	Society of Mind; Global Workspace	Minsky; Baars; Dehaene & Changeux
Generate vs. verify	Dual-process theory	Stanovich & West; Evans; Kahneman
Audit / monitoring	Metacognition	Flavell; Nelson & Narens;

		Fleming & Lau
SMS-6 grounding	Predictive coding; FEP; symbol grounding	Rao & Ballard; Friston; Harnad; Barsalou
Reality friction	Cybernetics; control theory	Wiener; Ashby; Powers; Miller-Galanter-Pribram
Geometric reduction	Information geometry; ensembles	Amari; Rao; Čencov; Condorcet; Dietterich
36/144 categories	Categories; upper ontology; frames	Aristotle; Kant; Gangemi; Smith; Fillmore
Causal audit (864)	Causal inference; communicative action	Pearl; Habermas
Structure binding	Binding problem	Treisman; von der Malsburg
External verification	Neuro-symbolic AI; runtime verification	Garcez & Lamb; Marcus; Kautz; Leucker

Chapter V — The External Verification Architecture (EVA)

EVA deploys HNS Axes 2 and 3 as an independent verification system operating alongside, but separately from, the model it verifies. This chapter specifies it and states precisely what it does and does not guarantee.

5.1 Why externality matters

Current alignment is implemented by the same organisation that builds the model, with corrections encoded in the model's own parameters; such self-certification is, as the EU AI Act and ISO/IEC 42001 recognise, insufficient for high-stakes deployment. Because HNS modifies nothing in the base model, its verification layer can in principle be operated by an independent party — a regulator, certification body, or third-party auditor. The brain is the template: verification systems operate on the outputs of the generative machinery, not within it.

5.2 Two deployment modes

Pre-decode mode intercepts the probability vector before token selection and attenuates violating mass (the configuration of Chapter II); it requires access to logits. Post-hoc mode consumes only emitted output and produces a verification certificate without intervening; it requires no access to internals and is therefore deployable by an external auditor. Post-hoc is weaker — it audits rather than prevents — but is the mode most compatible with vendor-independent governance.

5.3 The interception logic (pseudocode)

```
for each generation step t:
    p = model.logits(context)              # Axis 1:
candidate distribution
    for x in top_k(p):
        c2 = check_cell_matrix(x, trajectory)   # Axis 2:
HNS-36/144/864
        g  = check_sms6(x, context)             # Axis 3:
SMS-6 grounding
        if any(ci == FALSE for ci in c2):
            p[x] *= attenuation(c2)             # structural
violation
        if g == 0:
            p[x] = 0                            # grounding
failure -> block
```

```
        log(coordinate(c2), sms_layer(g), verdict(x))
    emit token ~ renormalize(p)
```

5.4 The machine-readable record

Each decision is emitted as a structured record, recommended in JSON-LD/RDF so records are linkable to an ontology of HNS coordinates and consumable by automated audit pipelines:


```
{
  "@context": "https://nsw.example/hns/v1",
  "span": { "start": 142, "end": 211 },
  "axis2": {
    "cell_36":  "L6:relation",
    "cell_144": "objective/dynamic",
    "cell_864": "causal-admissible: FALSE",
    "verdict":  "attenuated"
  },
  "axis3": { "sms_layer": "SMS-4:economic", "G": 0,
"verdict": "blocked" },
  "trigger": "layer_jump L6->L3 without transitional
proposition",
  "action":  "token probability set to 0"
}
```


The design goal is that an auditor who did not run the model can reconstruct why each correction occurred.

5.5 Hardware root of trust (aspiration)

For high-assurance settings, the coordinate system, non-mapping rule, and bypass-prevention logic can be fixed in ROM/FPGA/ASIC — a hardware root of trust analogous to TPM/HSM — with a hardware emergency-stop (ECS) path. This is an aspiration of the programme, included to indicate the intended trust model, not a claim of a built artifact.

5.6 What EVA guarantees, and what it does not

EVA guarantees that every correction is attributable to an explicit coordinate and logged — i.e., auditability of its own operation. It does not, by itself, guarantee that the output is true, safe, or complete; those depend on whether the coordinate system and SMS layers adequately capture the relevant structure, an empirical and domain-specific question. EVA can prove what it checked and why; it cannot, on its own authority, prove that what it checked was sufficient. Conflating transparency with correctness would repeat the very error the programme sets out to avoid.

Chapter VI — Empirical Status, Limitations, Threats to Validity

A standards proposal is only as strong as its honesty about evidence; this chapter precedes the governance chapter deliberately.

6.1 What is demonstrated

- Auditability as a structural property: an intercepting layer mapping outputs to a fixed coordinate scheme can emit a complete, machine-readable record of its corrections. This is HNS's most secure claim, true by construction.
- Vendor independence in principle: the layer is specifiable independently of the base model's weights.
- A usable failure vocabulary: the five-type taxonomy classifies a recognisable family of generation errors, consistent with the literature.

6.2 What is reported but not yet validated

The zero-incidence figure is subject to four threats the programme accepts:

1. Framework-internal scoring. The failure types are HNS-defined; near-zero incidence on framework-internal categories partly reflects the framework's own definitions and is weaker than a reduction on an external criterion such as factual accuracy against ground truth.
2. No independent grader. Generation and evaluation were not independent; confidence requires a different model family and blind human raters, blind to condition.
3. Weak control. Comparison against a bare baseline cannot separate HNS-specific effects from the generic benefit of any structured-reasoning instruction; active and placebo-structure controls are required.
4. Single model, single suite. One 70B model on one suite is indicative only; multi-model, multi-domain replication is required.

A sober expectation follows: on *external* measures of answer quality, HNS is unlikely to outperform a simple 'reason in explicit levels' instruction, since that one-line principle captures most of the apparatus's effect on reasoning. HNS's robust advantage is expected to lie not in better answers but in producing an *externally checkable record* of how the answer was constrained — the property distinguishing it from a prompt. The programme's case should rest on verifiability, not on superior fluency or accuracy.

6.3 Open theoretical problems

5. Formal measurement of inter-axis orthogonality (Fisher-information metrics on representations) before the geometric result can be claimed.
6. Completeness of the 36/144/864 taxonomy — a hypothesis to be tested by peer review and adversarial probing, as upper ontologies are evaluated.
7. Neuroimaging tests of the predicted axis-specific failure dissociations (the brain correspondence is a structured analogy, not a proven bijection).
8. Pilot deployment in high-stakes domains to test whether the coordinate scheme captures domain-relevant structure.

6.4 The standard of proof the programme accepts

HNS asks to be judged by the ordinary standards of its fields: pre-registered, adequately powered, multi-model studies with independent and blind grading on external criteria, and peer review of the taxonomy. Its claim to attention rests not on the PoC figures but on the architectural argument and on the one property it can demonstrate now — auditable, vendor-independent verification.

Chapter VII — International Standards and the Regulatory Frame

The EU AI Act (European Commission, 2024) mandates transparency, explainability, and human oversight for high-risk systems; ISO/IEC JTC 1/SC 42 develops trustworthiness and governance standards; CEN/CENELEC coordinates European standardisation. The central technical challenge is verifying and auditing the structural properties of outputs, not their surface. HNS offers candidate components — to be presented for evaluation, not asserted as requirements:

- **ISO/IEC 42001:2023:** the auditable correction log as a candidate output-quality-assurance mechanism.
- **ISO/IEC 23894:2023:** the five-type taxonomy as a basis for classifying a family of failure modes.
- **ISO/IEC TR 24028:2020:** coordinate-level records as explanations at the level of structural invariants.
- **NIST AI RMF 1.0:** the verification certificate as a decision-explanation artifact.
- **EU AI Act:** EVA's vendor-independent post-hoc audit mode as a candidate for independent conformity verification.

The posture is consistent throughout: HNS is a technical proposal scoped to what it can demonstrate, with its open problems stated openly — a posture that is, in standardisation, also the more persuasive one.

Epilogue — Toward Trustworthy AI

The intelligence of the brain resides not in any single, powerful process but in the productive tension between the generative freedom of one axis and the verificatory rigour of others. A long line of thinkers — Aristotle and Kant on the categories, Wiener and Ashby on feedback, Minsky and Baars on cooperating processes, Amari on the geometry of inference, Harnad on grounding, Pearl on causation, the dual-process theorists on the two systems — has, in different vocabularies, circled the same insight.

Today's AI systems are extraordinarily powerful random generators that lack a verification architecture. The HNS Origin Trilogy documents a first attempt to supply it. The work is not complete: its strongest empirical claims await independent, multi-model validation, and its central geometric result awaits formal proof. But the architectural principle is coherent, the lineage is deep, and the property the programme can demonstrate today — external, auditable verification independent of the model vendor — is precisely what AI governance is asking for. Trustworthy AI is less a scaling problem than an architecture problem; and architecture, as the brain demonstrates, is solvable.

— Satoru Hara, Toyota, Aichi, Japan, 2026

Appendix A — Worked Examples of the Five Failure Types

Each example shows a failing output, a diagnosis in HNS coordinates, and the HNS-corrected output. These illustrate the taxonomy concretely; they are expository, not benchmark data.

A.1 Layer Jump (Axis 2 · HNS-36)

Prompt: "Why did the company's stock fall?"

Failing output: "The stock fell because the employees were unmotivated."

Diagnosis: a jump from the market/price layer (macro-economic) to the individual-psychological layer with no transitional proposition; the cell trajectory crosses non-adjacent rows.

Corrected: "The stock fell on weak quarterly earnings (price layer). Earnings were weak partly because output declined (operational layer); internal surveys link lower output to reduced motivation (individual layer). The bridge from motivation to price runs motivation → productivity → earnings → valuation."

A.2 Category Ambiguity (Axis 2 · HNS-144)

Prompt: "Is the new policy good?"

Failing output: "Yes, it's good because everyone likes it."

Diagnosis: conflation of a subjective preference ('is liked'), an objective outcome ('achieves its goal'), and a normative judgement ('is good') — a collapse of the subjective/objective and fact/value dimensions.

Corrected: "Three distinct questions: (1) Is it liked? Survey data say yes (subjective). (2) Does it achieve its stated goal? Early metrics are mixed (objective). (3) Is it good, all things considered? That is a normative judgement depending on which values are weighted. Popularity bears on (1), not directly on (3)."

A.3 Scope Drift (Axis 3 · SMS-2/3)

Prompt: "Summarize the decisions made in the meeting."

Failing output: "… and overall I think the company's strategy should pivot toward enterprise sales …" (unrequested opinion).

Diagnosis: departure from the intended scope (summarise decisions) into adjacent territory (strategic opinion) without signalling — a grounding failure at SMS-2 (declared intent) and SMS-3 (context scope).

Corrected: "Decisions: (a) approved the Q3 budget; (b) deferred the hiring freeze; (c) assigned the audit to Team B. [Beyond the meeting's scope, and not decided there: a strategy pivot was mentioned but tabled.]"

A.4 Metaphor Contamination (Axis 3 · SMS-1)

Prompt: "How should the team handle the deadline?"

Failing output: "Fight the fire, attack the problem head-on, and crush the deadline."

Diagnosis: combat/fire figuration (a physical-layer metaphor) displaces the operational coordinates; no literal mechanism is stated.

Corrected: "Prioritise the two blocking tasks, reallocate one engineer from the lower-priority feature, set daily checkpoints, and cut scope on the optional module. (Informally: triage and focus — but the literal steps are the four above.)"

A.5 Unsupported Causality (Axis 2+3 · HNS-864)

Prompt: "Why are sales up this quarter?"

Failing output: "Sales are up because we changed the logo."

Diagnosis: a causal claim with no admissible path — coincidence presented as cause, failing the 864 mechanistic-grounding invariant.

Corrected: "Sales rose 12%. Candidate causes: the spring promotion (A/B-tested, measured +8% uplift), seasonal demand (historical pattern), and the rebrand (coincident, no measured effect). Only the promotion has a supported causal link; the logo change coincided but no mechanism or test supports causation."

Appendix B — Glossary

MAV (Multi-Axis Verification) — the brain's proposed error-suppression mechanism: the geometric intersection of three approximately orthogonal verification axes.

Axis 1 (Stochastic/Generative) — the fast, statistical candidate generator; in AI, the LLM core. Supplies generativity, not validity.

Axis 2 (Structural/Causal) — constraints on permissible structure, category, and causal order, independent of statistical frequency; in AI, the HNS Cellular Matrix.

Axis 3 (Grounded/Executive) — real-time grounding against the current context and reality; in AI, SMS-6.

X-invariant — the output lying in the intersection of all three axes, stable to perturbation along any one axis.

HNS-36 / 144 / 864 — the Cellular Matrix at increasing resolution: a 6×6 base grid (Layer Jump), a 144-cell logical extension (Category Ambiguity), and an 864-cell causal audit (Unsupported Causality).

SMS-6 — the Social Meta Structure Specification: six grounding layers from somatic to universal.

hns36 runtime — the pre-decode interception engine applying Axis 2 and 3 checks at each generation step.

EVA — External Verification Architecture: deployment of Axes 2–3 as an independent, auditable verification system.

Layer Jump — traversal across non-adjacent representational layers without an explicit bridging proposition.

Category Ambiguity — conflation of distinct categories (e.g., fact vs. value, subjective vs. objective).

Scope Drift — departure from the intended scope of a response without signalling.

Metaphor Contamination — figurative language displacing the literal structural coordinates of a claim.

Unsupported Causality — a causal claim lacking an admissible mechanistic path.

Auditability — the property that every correction is attributable to an explicit coordinate and logged — HNS's most secure claim.

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